SSCO AI HACKATHON · STERN STEWART & CO. · 21–22 OCT 2026

Expansion Navigator: a screenshot journey

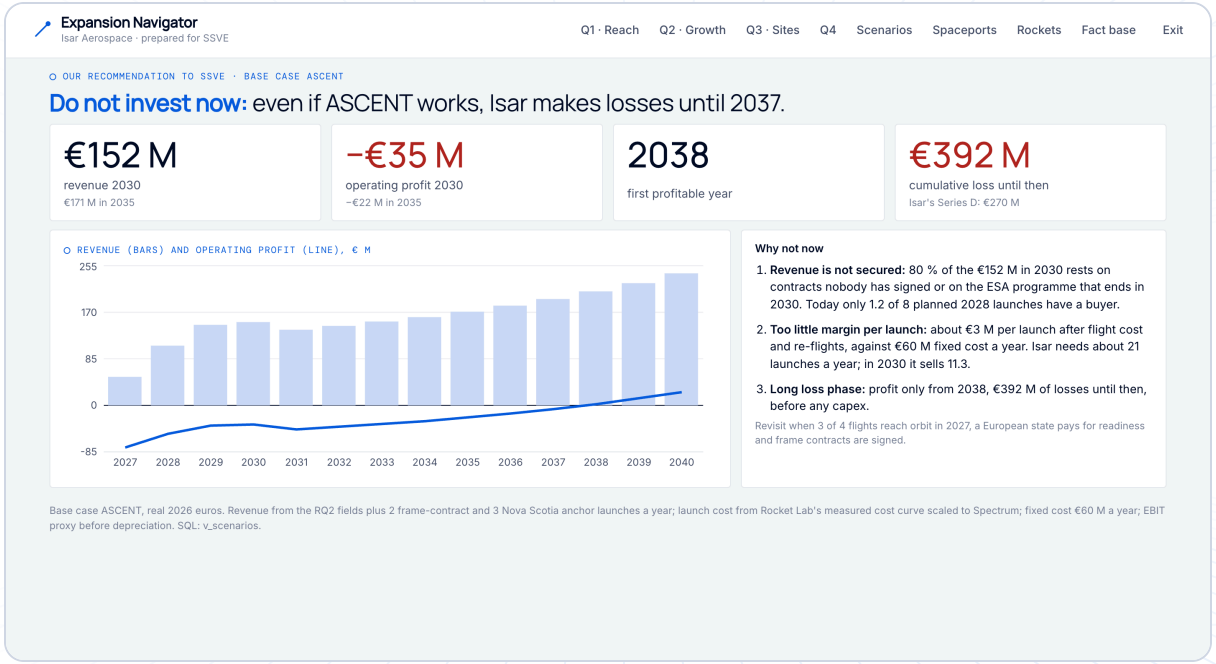
Our decision tool for the Isar Aerospace case, screen by screen.

1 **The tool, screen by screen**
22 screenshots of the final build, clickable like the tool
pp. 2–24 →

2 **The case behind the screens**
Our recommendation to SSVE, in consulting format
pp. 25–29 →

Read straight through, or click like in the tool.

Benedikt Supper · Luca Marie Repa · Nils Wagner



Case exercise for Stern Stewart & Co. Team estimates, not Isar data or investment advice. Contains participant-only material.

PART 1

The tool, screen by screen

22 screenshots of the final build, in the order we clicked through it. Turn the pages, or click the navigation, tiers and toggles as in the tool.

1 ▶ Present: the answer p. 3

2 Q1 · Reach p. 4

3 Q2 · Growth p. 8

4 Q3 · Sites p. 11

5 Q4 · Shock p. 15

6 Scenarios p. 16

7 Spaceports p. 21

8 Rockets p. 23

9 Fact base p. 24

OUR RECOMMENDATION TO SSVE · BASE CASE ASCENT

Do not invest now: even if ASCENT works, Isar makes losses until 2037.

Explained · p. 27

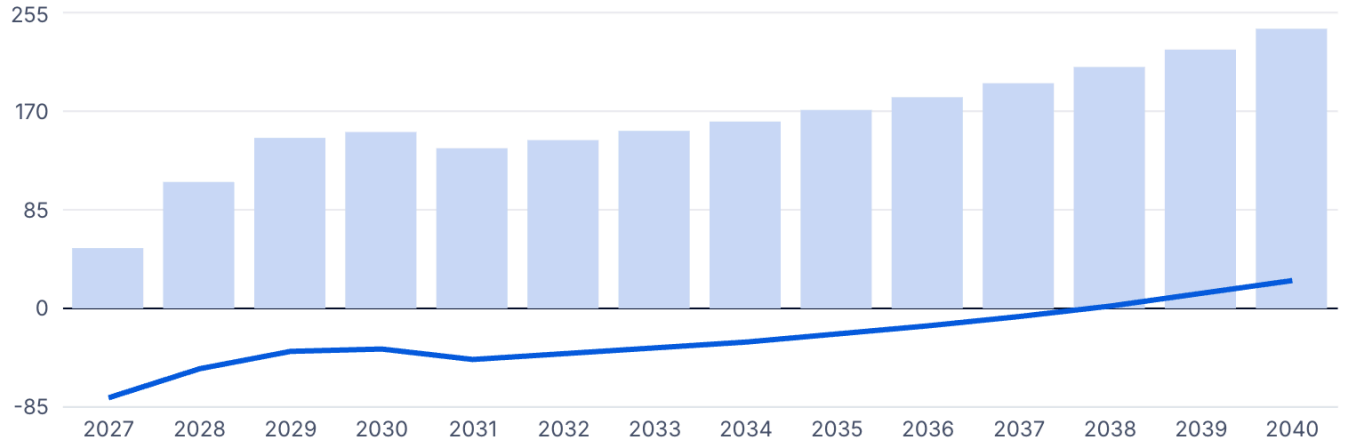
€152 M
revenue 2030
€171 M in 2035

−€35 M
operating profit 2030
−€22 M in 2035

2038
first profitable year

€392 M
cumulative loss until then
Isar's Series D: €270 M

REVENUE (BARS) AND OPERATING PROFIT (LINE), € M



Why not now

1. **Revenue is not secured:** 80 % of the €152 M in 2030 rests on contracts nobody has signed or on the ESA programme that ends in 2030. Today only 1.2 of 8 planned 2028 launches have a buyer.
2. **Too little margin per launch:** about €3 M per launch after flight cost and re-flights, against €60 M fixed cost a year. Isar needs about 21 launches a year; in 2030 it sells 11.3.
3. **Long loss phase:** profit only from 2038, €392 M of losses until then, before any capex.

Explained · p. 26

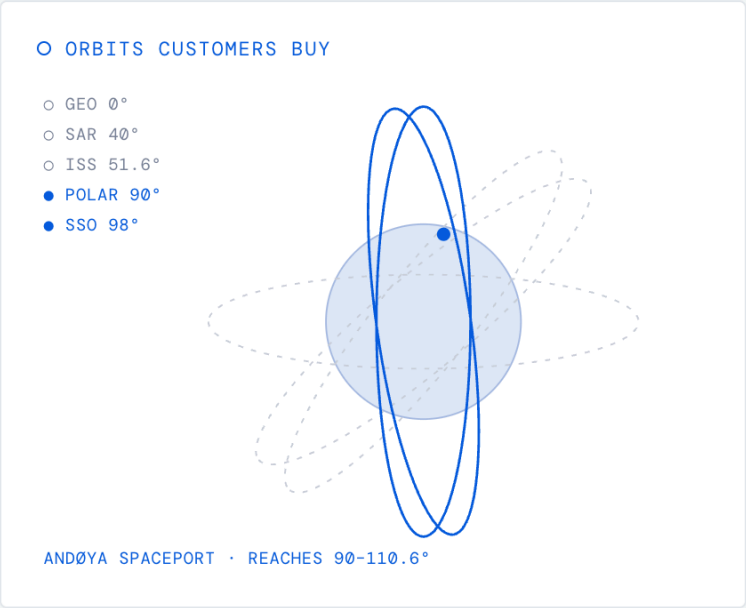
Revisit when 3 of 4 flights reach orbit in 2027, a European state pays for readiness and frame contracts are signed.

Base case ASCENT, real 2026 euros. Revenue from the RQ2 fields plus 2 frame-contract and 3 Nova Scotia anchor launches a year; launch cost from Rocket Lab's measured cost curve scaled to Spectrum; fixed cost €60 M a year; EBIT proxy before depreciation. SQL: v_scenarios.

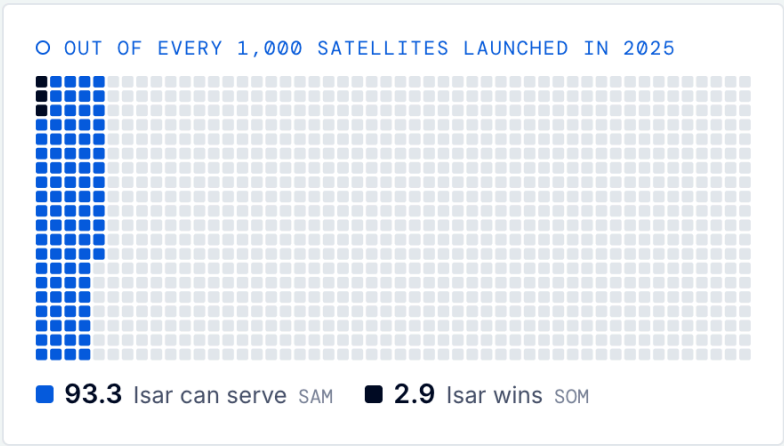
Re-checked after the final: the 80 % include the signed ESA programme (54 % without it); “1.2 of 8” is modelled from flown demand, not a count of bookings; “about 21 launches” uses the 2030 cost per flight (15 to 17 on the cost curve); the €270 M Series D is past funding, not available cash; with the Kourou fee break-even moves to 2039.

QUESTION 1 · TODAY

What share of global launch demand can Isar serve from Andøya?



From Andøya, Isar can serve **422** of the 4,521 satellites launched in 2025 (**9.3%**), but only 0.66% of their mass. It wins about **13 satellites** (563 kg, €10 M) a year: demand for dedicated launches, not capacity, sets the limit today.



CLICK A TIER TO SEE WHAT NARROWS IT

Satellites | Mass | Revenue

Assumptions Reset

CLICK A TIER TO SEE WHAT NARROWS IT

Satellites | Mass | Revenue

TAM
Global launch demand
4,521
satellites / yr
100%
Every satellite that reached orbit
Shown below

SAM
Serviceable from Andøya
422
satellites / yr
9.3% of TAM
Open to Isar, fits on Spectrum, orbit reachable
Open details

SOM
Obtainable by Isar
13
satellites / yr
0.29% of TAM
Buys a dedicated launch, chooses Isar, within its capacity
Open details

TAM · IN DETAIL

Source reliability ● high ● medium ● low

Global launch demand is measured, not estimated: every satellite that reached orbit in the base year.

Satellites	Mass	Launch revenue
4,521	2,694 t	€11.1 bn
● GCAT	● GCAT	● SIA 2026

Revenue: $12.4 bn global launch revenue, converted at 1.12 $/€. Launched satellites are realised demand, capped by SpaceX's capacity: a lower bound rather than a ceiling.

Four counts, one choice

2025	Value	Source	Why it differs

Who launched in 2025

Starlink	3,258 72.1%	2,043 t	
Commercial (on	442 9.8%	83.2 t	

Assumptions

Reset

TAM · WHAT COUNTS AS DEMAND

Base year of flown demand **2025**

2024 | 2025

● High confidence Nils ● GCAT

▶ Why 2025?

SAM · WHAT NARROWS IT

Launch sites **Andøya**

Andøya | + Nova Scotia | + Kourou

● High confidence Nils

● CAA Norway ● Isar · Nova Scotia ● CNES

▶ Why Andøya?

Other governments' demand open to Isar **20%**

0% — 100%

● Low confidence Benedikt ● Team estimate

▶ Why 20%?

SOM · WHAT ISAR WINS

Share of the serviceable market that buys a dedicated launch **7%**

4% — 15%

SAM · IN DETAIL

Source reliability ● high ● medium ● low

Three narrowing factors turn global demand into what Isar can serve from its launch sites.

○	**Start: TAM 2025**	4,521 sats · 2,694 t	
1	**Access: who may book Isar** Excludes Starlink, China, Russia, Kuiper/OneWeb and the US government; other governments count 20% ● GCAT ● NSPM-17 ● Team estimate	**506 sats** −88.8% **124 t** −95.4%	
2	**Payload: fits on Spectrum** Satellite weighs at most 700 kg in SSO or 1,000 kg in other low orbits (Spectrum's published payload) ● Isar · Spectrum	**483 sats** −4.5% **24.2 t** −80.5%	
3	**Orbit: the launch site reaches it** Inclination 90–110.6° (Andøya Spaceport) ● CAA Norway ● Andøya Space	**422 sats** −12.6% **17.9 t** −26.2%	

SAM revenue: 17.9 t × rideshare to dedicated price per kg (7,500–20,000 $/kg) = €120 M–319 M.

All satellites of 2025 by inclination: blue is reachable



● CAA Norway ● Isar · Nova Scotia ● CNES

▶ Why Andøya?

Other governments' demand open to Isar **20%**

0% 100%

● Low confidence Benedikt ● Team estimate

▶ Why 20%?

SOM · WHAT ISAR WINS

Share of the serviceable market that buys a dedicated launch **7%**

4% 15%

● Medium confidence Benedikt

● GCAT ● BryceTech 2026 ● Schingler (Planet)

▶ Why 7%?

Isar's share of the dedicated demand **45%**

20% 100%

● Low confidence Benedikt

● GCAT ● Schingler (Planet)

▶ Why 45%?

Spectrum launches per year **4 / yr**

1 / yr 12 / yr

● Medium confidence Luca

SOM · IN DETAIL

Source reliability ● high ● medium ● low

Three steps decide what Isar wins: who pays for a dedicated launch, how much of it chooses Isar over Rocket Lab, and how much Spectrum can fly.

Start: SAM — 422 sats · 17.9 t

1 Penetration: buys a dedicated launch

7% of the SAM pays for its own launch instead of rideshare

● GCAT ● BryceTech 2026 ● Schingler (Planet)

30 sats −93%

1.25 t −93%

2 Win share: chooses Isar

45% of that dedicated demand flies Spectrum instead of Rocket Lab's Electron or a later entrant

● GCAT ● Schingler (Planet)

13 sats −55%

563 kg −55%

3 Capacity: Isar can fly it

4 launches × 75% success × 700 kg × 100% fill = 2.1 t

● Payload ● Isar · missions ● CAA Norway ● Schingler (Planet) ● Team estimate

13 sats not binding

563 kg not binding

SOM revenue: 0.8 Spectrum loads × €12 M = €10 M.

Demand meets capacity

Buys a dedicated launch	1.25 t
... of which chooses Isar	563 kg
Isar can fly	2.1 t

The SOM is the smaller of what chooses Isar and what Isar can fly: 563 kg, about 0.8 Spectrum loads. Today demand sets the limit. With 75% success Isar

Who flies Isar's market today

Falcon 9	366	86.8%	14.6 t
Electron	28	6.7%	751 kg
Nuri	11	2.6%	570 kg
Soyuz-2-1B	7	1.7%	114 kg
Vega C	5	1.2%	1.32 t
Lijian-1	5	1.1%	386 kg

Assumptions

Reset

TAM · WHAT COUNTS AS DEMAND

Base year of flown demand: **2025**

2024 | **2025**

● High confidence Nils ● GCAT

▶ Why 2025?

SAM · WHAT NARROWS IT

Launch sites: **Andøya**

Andøya | + Nova Scotia | + Kourou

● High confidence Nils

● CAA Norway ● Isar · Nova Scotia ● CNES

▶ Why Andøya?

Other governments' demand open to Isar: **20%**

0% — 100%

● Low confidence Benedikt ● Team estimate

▶ Why 20%?

SOM · WHAT ISAR WINS

Share of the serviceable market that buys a dedicated launch: **7%**

4% — 15%

Isar should stay a launch provider and grow through **contracts**, not rockets on stock. In 2030 the fields we recommend bring about **€88.1 M** revenue, the equivalent of 7.3 Spectrum launches, and an EVA proxy of **+€28.8 M**. Today's flown demand fills only 2.7 of the 8 launches planned for 2028, of which Isar can win about 1.2.

The launch gap: planned launches against visible demand



Dark blue: launches Isar can realistically win from today's flown demand (45 % against Rocket Lab). Light blue: all visible dedicated demand. Outline: launches Isar plans. Everything beyond the light bar needs contracts. SQL: v_q2_launch_gap.

STRATEGY ASCENT: ANCHOR CONTRACTS FIRST, THEN ROCKETS

A
Anchor contracts first

Frame contracts over several launches with deposits. The launch count follows contracts, not factory capacity.

`Subscription, Lock-in`

S
Sovereign readiness

States pay for readiness: reserved slots, a rocket on stock, lost satellites replaced within days together with Planet.

`Guaranteed Availability`

C
Core polar launches

Dedicated launches to polar and sun-synchronous orbits from Andøya for commercial, institutional and defence customers.

`Aikido: the opposite of rideshare`

E
Extras on every launch

Integration, insurance brokerage, drop-off in several orbits, selling spare capacity.

`Add-on`

N
No-go list

No satellites, no own constellation, no transfer vehicle, no price war with rideshare.

`Not: Integrator, License, No Frills`

T
Triggers for expansion

Nova Scotia, Kourou, hypersonic tests and a mid-class rocket only with an anchor customer, funding or proven cadence.

`EVA discipline`

EXPANSION FIELDS IN NUMBERS

2030 | 2035 | 2040

EXPANSION FIELDS IN NUMBERS

2030 | 2035 | 2040

Growth of paid readiness **15% a year**

No European budget is published; the US programme is the proxy. Range 10–25 %.

Cost of capital (WACC) **12%**

Charged on capital a field ties up, from 2027 on.

Field	Market 2025	Isar revenue 2030	Gross profit	EVA proxy 2030		Recommendation
European institutional launches (ESA European Launcher Challenge) Core launch · Andøya	€180.4 M	€39.5 M	€16.1 M		**+€16.1 M**	Pursue
Dedicated launches to polar and sun-synchronous orbits (Andøya) Core launch · Andøya	€21.5 M	€14.9 M	€6.1 M		**+€6.1 M**	Pursue
More customers buy their own launch (rideshare bottleneck after 2028) Core launch · Andøya	€15.3 M	€10.6 M	€4.3 M		**+€4.3 M**	Pursue
Suborbital hypersonic test flights New product · Andøya	€10.6 M	€8.5 M	€3.5 M		**+€2.3 M**	With trigger
Paid launch readiness for European states New product · Andøya (second pad or reserved slots)	€26.8 M	€21.6 M	€8.8 M		**+€1.4 M**	Pursue
Building satellites (Rocket Lab model) Beyond launch · Parsdorf	€18.2 bn	€23.2 M	€7.3 M		**+€1.3 M**	Deliberately not
Extras on every launch (integration, insurance brokerage, multi-orbit drop-off) New product · all	€2.1 M	€1.5 M	€0.9 M		**+€0.9 M**	Pursue
Selling engines to other launchers Beyond launch · Parsdorf	€7.3 bn	€5.3 M	€1.6 M		**−€0.8 M**	Deliberately not

Building satellites (Rocket Lab model) Beyond launch · Parsdorf	€18.2 bn	€23.2 M	€7.3 M		**+€1.3 M**	Deliberately not
Extras on every launch (integration, insurance brokerage, multi-orbit drop-off) New product · all	€2.1 M	€1.5 M	€0.9 M		**+€0.9 M**	Pursue
Selling engines to other launchers Beyond launch · Parsdorf	€7.3 bn	€5.3 M	€1.6 M		**−€0.8 M**	Deliberately not
Own orbital transfer vehicle Beyond launch · Parsdorf	€1.7 bn	€15.7 M	€4.7 M		**−€2.5 M**	Deliberately not
Low inclinations and GTO from Kourou New site · Kourou	€3.5 M	€1.1 M	€0.4 M		**−€4.6 M**	With trigger
Own constellation and data services Beyond launch · –	€5.2 bn	€14.5 M	€8.1 M		**−€9.9 M**	Deliberately not
Mid-inclination orbits 45–90° from Nova Scotia New site · Nova Scotia	€6 M	€4.1 M	€1.7 M		**−€11.7 M**	With trigger
Medium-lift rocket (2–20 t, partly reusable) New product · Kourou or a new site	€1.1 bn	–	–		**−€48 M**	With trigger
All “Pursue” fields		€88.1 M	€36.2 M	**+€28.8 M**		
All “With trigger” fields		€13.7 M	€5.6 M	**−€62 M**		
All “Deliberately not” fields		€58.7 M	€21.7 M	**−€11.9 M**		

EVA proxy = gross profit − fixed cost − WACC × capital tied up. Gross profit stands in for operating profit after tax, so it ranks the fields; it is not a valuation. Click a row for the calculation and its sources. SQL: v_q2_expansion_fields.

Source reliability ● high ● medium ● low

QUESTION 3 · SITE PORTFOLIO

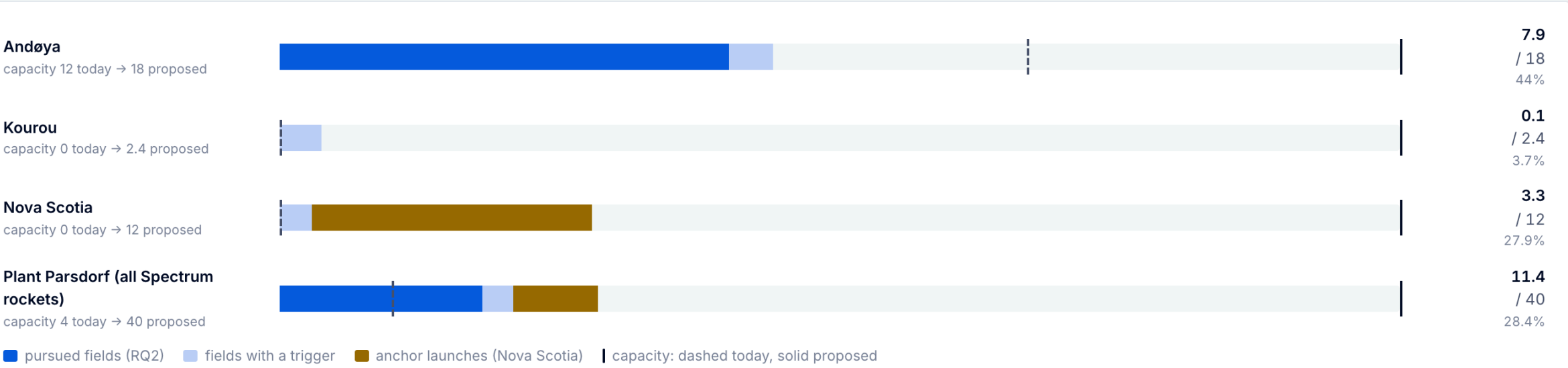
How should Isar structure its production and launch sites?

Explained · p. 26

One plant, one core spaceport with a second pad, one sovereign site abroad. Production stays in Bavaria; launch sites follow customers and contracts. In 2030 Isar flies about **11.4 launches**: 7.9 from Andøya (44% of its proposed capacity), 3.3 from Nova Scotia. The plant runs at **28.4%** of its 40 rockets a year, so a second plant is not needed.

LAUNCHES EACH SITE HAS TO CARRY

2030 | 2035 | 2040



Canada: build rockets there or ship them?

Rockets a year a Canadian line would build **3**

A second line building 3 of 11.4 rockets a year costs about **2× as much per rocket** as one plant building all of them (cost falls to 70% with every doubling of annual output, as at Rocket Lab 2023–2025). Shipping stages across the Atlantic and integrating them at the pad keeps the scale in Bavaria.

A Canadian plant only makes sense if a Canadian anchor contract demands local value that integration, launch operations and Canadian suppliers cannot deliver, and volume exceeds about 10 rockets a year.

Questions for Isar's management

1. Who is the anchor customer at Nova Scotia, and when does it sign?
2. Where should up to 40 launches a year at Nova Scotia come from while US-built satellites cannot launch from Canada?
3. Who uses Andøya's second pad, and does Isar get reserved slots for readiness contracts?
4. Production or only final integration in Canada, and how much local value does the customer require?
5. From which contracted cadence would Isar build a second plant?

THE PORTFOLIO TODAY AND AS PROPOSED

PRODUCTION · VATERSTETTEN (PARSDORF), BAVARIA — Expand

Serial production plant

TODAY

Plant of more than 40,000 m², nearing completion in Feb 2026; builds Spectrum (vehicles for flights 3 to 7)

PROPOSED

The single serial plant for every launch site;

TESTING · ESRANGE, KIRUNA — Keep

Engine and stage test site

TODAY

Engine tests since 2020; second test site since Feb 2026 with more than 30 engine tests a month and full-stage acceptance

PROPOSED

Keep as the acceptance step for every stage;

LAUNCH · ANDØYA, NORDLAND — Expand

Andøya, second pad

TODAY

In Norway's 2027 budget proposal (about NOK 646 M), state-owned, tenant not named

PROPOSED

Readiness and redundancy: reserved slots for paid readiness and a fallback if pad A is

LAUNCH · ANDØYA, NORDLAND — Keep

Andøya, pad A (exclusive)

TODAY

Isar's only orbital pad: flight 1 in March 2025, flight 2 reached orbit in September 2026

PROPOSED

Core site for polar and sun-synchronous launches

PRODUCTION · VATERSTETTEN (PARSDORF), BAVARIA — Expand

Serial production plant

TODAY

Plant of more than 40,000 m², nearing completion in Feb 2026; builds Spectrum (vehicles for flights 3 to 7)

PROPOSED

The single serial plant for every launch site; stages are shipped, final integration happens at the pad

TRIGGER

A second plant only above about 30 contracted launches a year, or if a Canadian local-content duty cannot be met by integration, launch operations and suppliers

- Medium confidence
- Isar/VGP PR 2024
- Isar · Series D
- Isar · missions
- Payload

TESTING · ESRANGE, KIRUNA — Keep

Engine and stage test site

TODAY

Engine tests since 2020; second test site since Feb 2026 with more than 30 engine tests a month and full-stage acceptance

PROPOSED

Keep as the acceptance step for every stage; no Isar orbital launches here (Firefly is the anchor tenant)

- Medium confidence
- Isar Esrange PR 2026
- SSC

LAUNCH · ANDØYA, NORDLAND — Expand

Andøya, second pad

TODAY

In Norway's 2027 budget proposal (about NOK 646 M), state-owned, tenant not named

PROPOSED

Readiness and redundancy: reserved slots for paid readiness and a fallback if pad A is blocked

TRIGGER

A readiness contract with a European state, or more than 10 Isar launches a year at Andøya

- Medium confidence
- Space Launch Schedule
- TU.no 2026
- KeepTrack 10/2026
- Team estimate

LAUNCH · ANDØYA, NORDLAND — Keep

Andøya, pad A (exclusive)

TODAY

Isar's only orbital pad: flight 1 in March 2025, flight 2 reached orbit in September 2026

PROPOSED

Core site for polar and sun-synchronous launches

- High confidence
- Isar · pad lease
- CAA Norway
- ESF 2025 (stages)

LAUNCH · KOUROU, FRENCH GUIANA — Option

Guiana Space Centre, ELM-Diamant

TODAY

One of five laureates of the shared Diamant complex (term sheet 2022); no build date

PROPOSED

Option for low inclinations and GTO, only with a European anchor mission

TRIGGER

An EU or ESA anchor mission and a build date

LAUNCH · CANSO, NOVA SCOTIA — Option

Spaceport Nova Scotia, Isar complex

TODAY

Dedicated Spectrum complex signed July 2026 (Isar Aerospace Canada Inc.): pad handover 1 Nov 2026, first launch 2028, Maritime Launch Services provides assembly halls; 3.75 M USD per quarter after a 30-month waiver. Isar names up to 40 launches a year by 2029

PROPOSED

HEADQUARTERS · OTTOBRUNN, BAVARIA — Keep

Headquarters and development

TODAY

Headquarters, engineering, development of Spectrum and upgrades

PROPOSED

Unchanged: one engineering centre for all sites

- High confidence
- Isar · Series D

PRODUCTION · NOVA SCOTIA — Not now

Production plant in Canada

TODAY

–

PROPOSED

Not recommended: at 1–3 rockets a year a second line costs about twice as much per rocket

TRIGGER

Only if a Canadian defence contract (ITB: 100 % Canadian activity above CAD 100 M) cannot

● Medium confidence

● Isar/VGP PR 2024 ● Isar · Series D
● Isar · missions ● Payload

LAUNCH · KOUROU, FRENCH GUIANA — Option

Guiana Space Centre, ELM-Diamant

TODAY

One of five laureates of the shared Diamant complex (term sheet 2022); no build date

PROPOSED

Option for low inclinations and GTO, only with a European anchor mission

TRIGGER

An EU or ESA anchor mission and a build date

● Low confidence

● CNES ● AdvTV 2024 (Canopée)
● Team estimate

LAUNCH · CANSO, NOVA SCOTIA — Option

Spaceport Nova Scotia, Isar complex

TODAY

Dedicated Spectrum complex signed July 2026 (Isar Aerospace Canada Inc.): pad handover 1 Nov 2026, first launch 2028, Maritime Launch Services provides assembly halls; 3.75 M USD per quarter after a 30-month waiver. Isar names up to 40 launches a year by 2029

PROPOSED

Sovereign site for Canada and NATO and mid inclinations (45–98°); rockets shipped from Bavaria, integrated on site, no Canadian plant

TRIGGER

At least 3 anchor launches a year (Canadian DND or NATO) by the end of the fee waiver, about 2030

● Medium confidence

● Isar/MLS PR 2026 ● Isar · Nova Scotia
● SpaceQ MLS Q2 2026 ● SpaceQ TSA 2026
● ISED ITB backgrounder ● BetaKit 2026 (DND)

HEADQUARTERS · OTTOBRUNN, BAVARIA — Keep

Headquarters and development

TODAY

Headquarters, engineering, development of Spectrum and upgrades

PROPOSED

Unchanged: one engineering centre for all sites

● High confidence ● Isar · Series D

PRODUCTION · NOVA SCOTIA — Not now

Production plant in Canada

TODAY

–

PROPOSED

Not recommended: at 1–3 rockets a year a second line costs about twice as much per rocket

TRIGGER

Only if a Canadian defence contract (ITB: 100 % Canadian activity above CAD 100 M) cannot be met by integration, launch operations and Canadian suppliers, and volume exceeds about 10 rockets a year

● Low confidence

● Rocket Lab 10-K ● ISED ITB backgrounder
● OAG ITB 2024 ● Team estimate

Q4 · The shock. Norway closes the sea and air space around Andøya; Canada reserves Nova Scotia for Canadian payloads. Shown: the shock lever set to a full year and its result. From the build of 22 Oct, 14:04; the rest of that page is omitted.

FIVE ASSUMPTIONS TO TEST · ASCENT (BASE)

- Frame-contract launches a year **2** base 2
- Paid readiness grows **15% a year** base 15%
- Success rate from 2030 **92%** base 92%
- Fixed cost base (all scenarios) **€60 M a year** base €60 M
- Q4 shock: Andøya closed in 2030 **12 months** base 0

Explained · p. 27

Q4 shock · Andøya closed, Nova Scotia only Canadian

24.6%	46.5%	−€64 M
revenue kept, Andøya and Nova Scotia only	with a Kourou pad share	EBIT 2030 (base −€35 M)

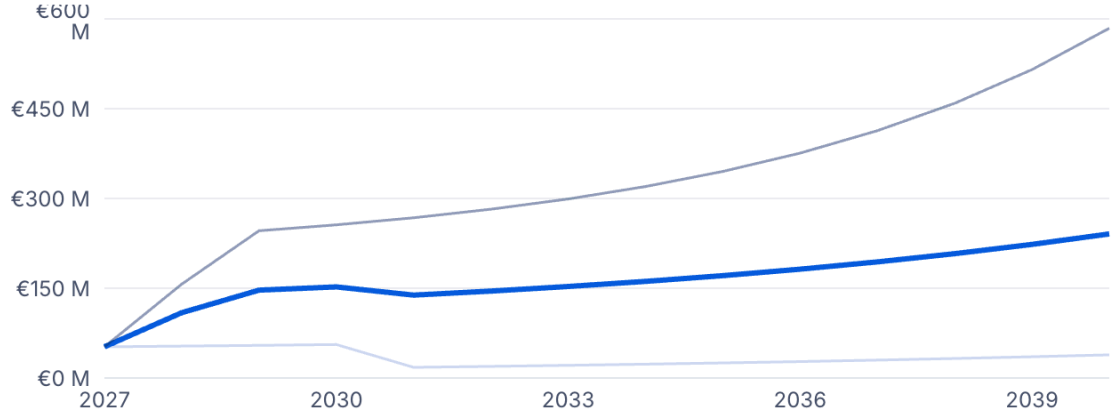
Second ocean: production is not hit, the launch site is. Hold a Kourou pad share as insurance (2.4 launches, about €5 M a year), let Nova Scotia fly Canadian payloads, and keep a fallback site in every readiness contract.

ASCENT (base): profitable from 2038. Revenue €152 M in 2030 and €171 M in 2035; the cumulative operating loss bottoms out at €392 M in 2037. Our strategy executed: contracts first, Nova Scotia only with an anchor.

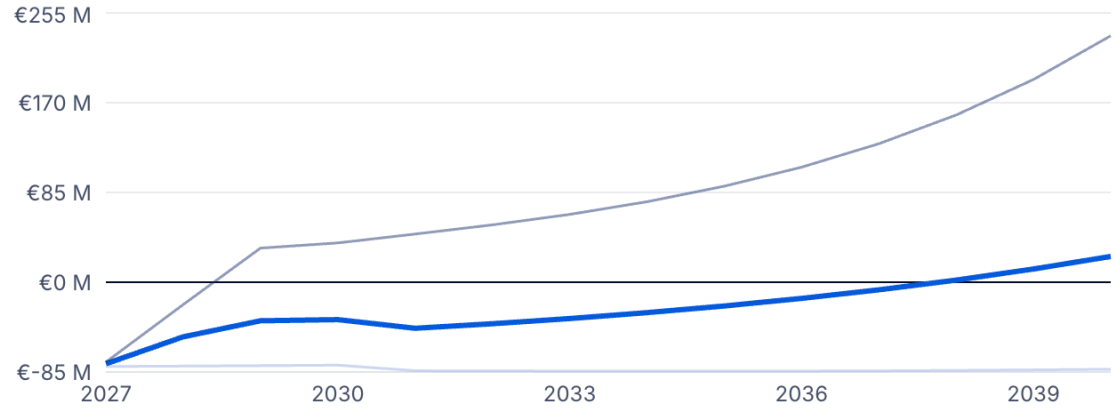
Bear | **ASCENT (base)** | Bull

Explained · p. 27

REVENUE PER YEAR



OPERATING PROFIT (EBIT PROXY) PER YEAR



Frame-contract launches a year from 2029 **2**

The A in ASCENT: launches sold beyond today's visible demand. Base: 2.

Fixed cost base a year **€60 M**

The least certain input: R&D and overhead. Base €60 M.

Spectrum cost per launch vs Electron **1.5×**

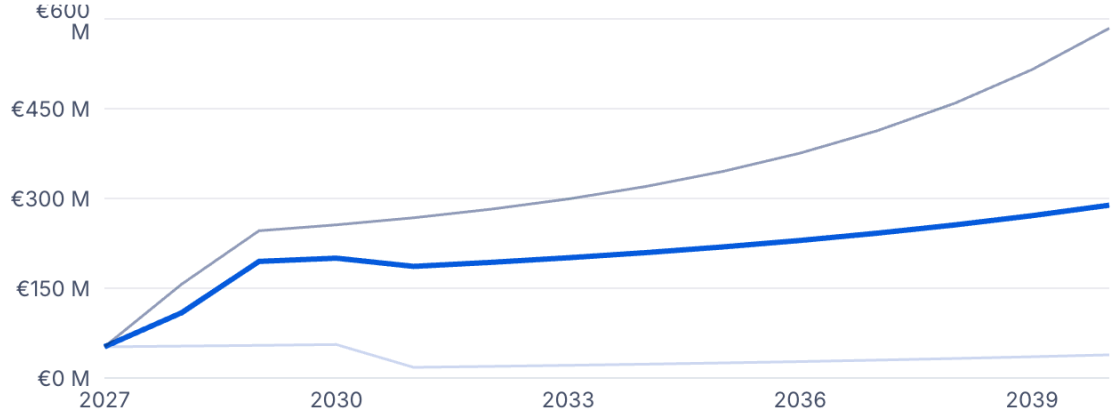
Spectrum carries about 3× Electron's payload with the same engine count. Base 1.5×.

ASCENT (base): profitable from 2034. Revenue €200 M in 2030 and €219 M in 2035; the cumulative operating loss bottoms out at €163 M in 2033. Our strategy executed: contracts first, Nova Scotia only with an anchor.

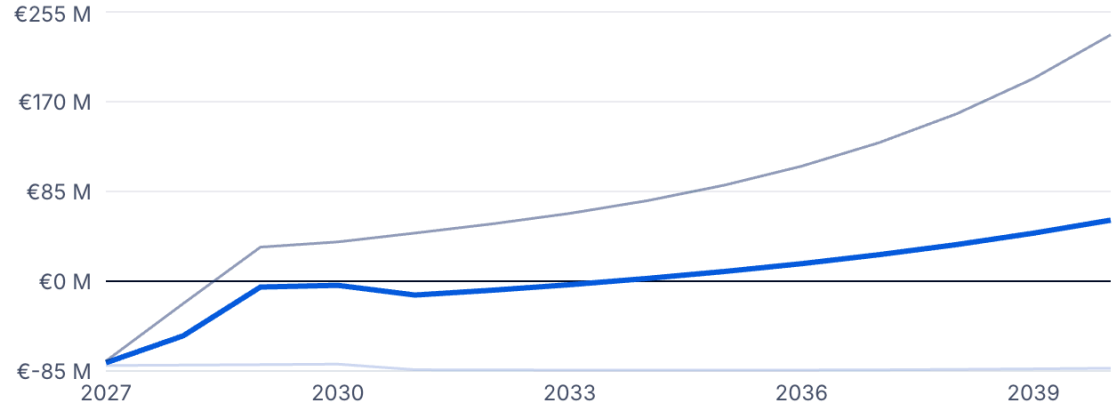
Bear | **ASCENT (base)** | Bull

Back to base

REVENUE PER YEAR



OPERATING PROFIT (EBIT PROXY) PER YEAR



Explained · p. 27

Frame-contract launches a year from 2029 **6**

The A in ASCENT: launches sold beyond today's visible demand. Base: 2.

Fixed cost base a year **€60 M**

The least certain input: R&D and overhead. Base €60 M.

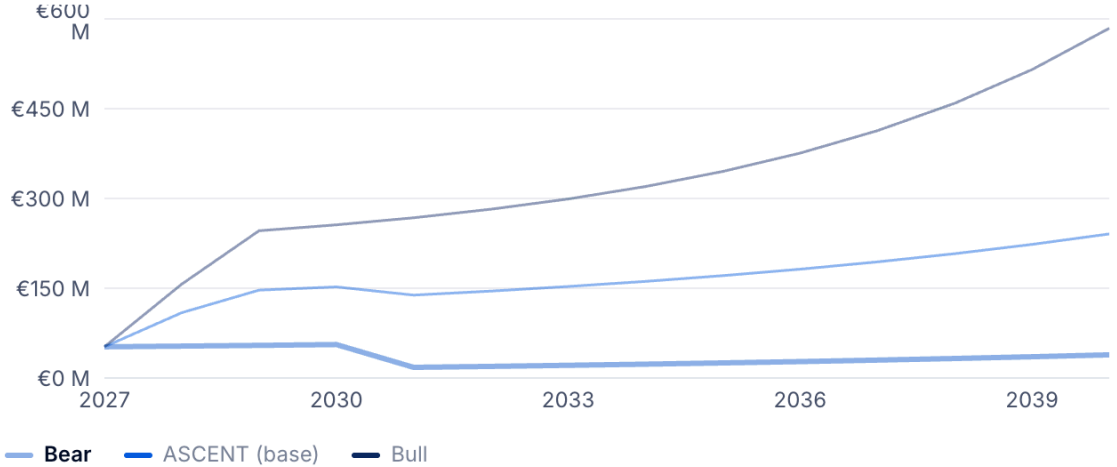
Spectrum cost per launch vs Electron **1.5×**

Spectrum carries about 3× Electron's payload with the same engine count. Base 1.5×.

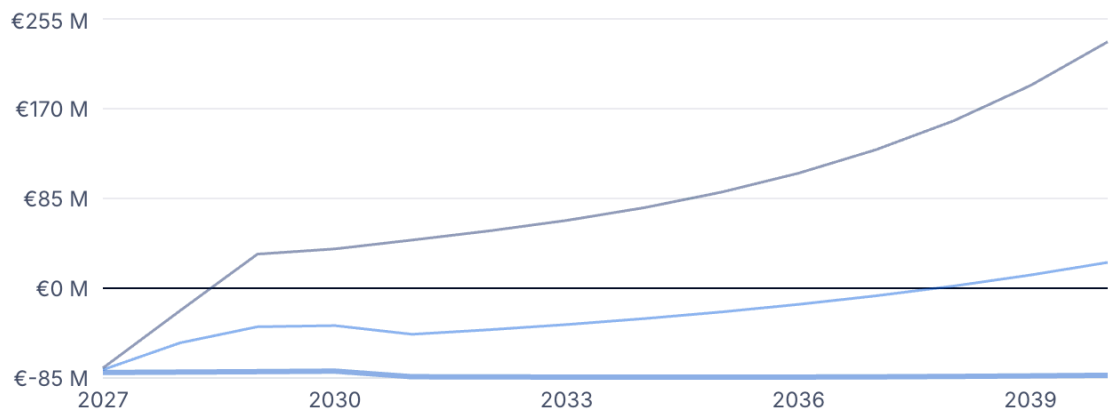
Bear: no operating profit by 2040. Revenue €56 M in 2030 and €25 M in 2035; the cumulative operating loss bottoms out at €1,154 M in 2040. ASCENT does not land: no contracts beyond today's demand, reliability stalls.

Bear | ASCENT (base) | Bull

REVENUE PER YEAR



OPERATING PROFIT (EBIT PROXY) PER YEAR



Frame-contract launches a year from 2029 **0**

The A in ASCENT: launches sold beyond today's visible demand. Base: 0.

Fixed cost base a year **€60 M**

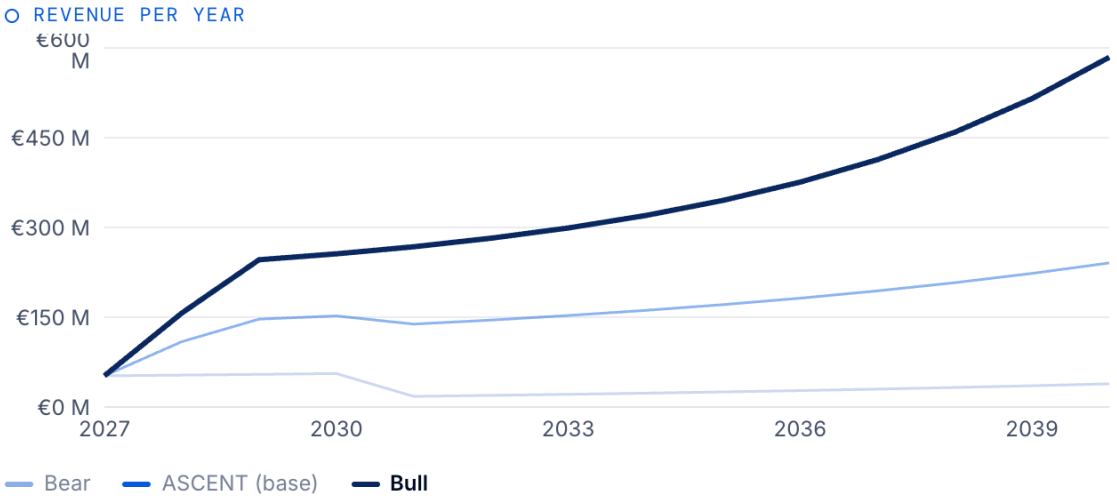
The least certain input: R&D and overhead. Base €60 M.

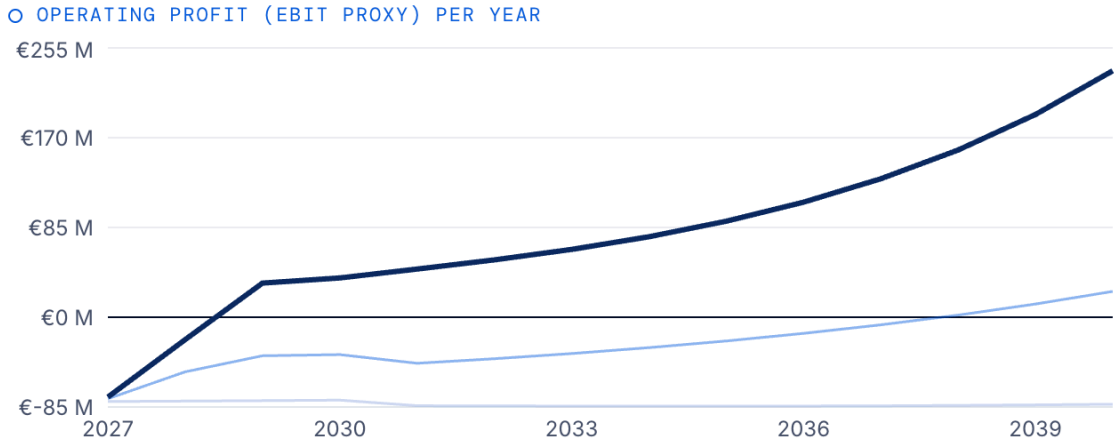
Spectrum cost per launch vs Electron **1.5×**

Spectrum carries about 3× Electron's payload with the same engine count. Base 1.5×.

Bull: profitable from 2029. Revenue €256 M in 2030 and €345 M in 2035; the cumulative operating loss bottoms out at €97 M in 2028. Sovereign demand surges: Europe buys readiness and launches at scale.

Bear | ASCENT (base) | **Bull**





Frame-contract launches a year from 2029 **6**

The A in ASCENT: launches sold beyond today's visible demand. Base: 6.

Fixed cost base a year **€60 M**

The least certain input: R&D and overhead. Base €60 M.

Spectrum cost per launch vs Electron **1.5×**

Spectrum carries about 3× Electron's payload with the same engine count. Base 1.5×.

P&L of ASCENT (base), € M (real 2026 euros)

	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
Launches sold	4.2	8	11	11.3	10	10.3	10.7	11.1	11.5	12	12.6	13.2	13.9	14.7
Flights incl. re-flights	5.2	9.5	12.5	12.3	10.8	11.2	11.6	12	12.5	13.1	13.7	14.4	15.1	15.9
Success rate	81%	84.7%	88.3%	92%	92%	92%	92%	92%	92%	92%	92%	92%	92%	92%
Cost per flight	13.1	9.7	8.4	8.5	9	8.9	8.7	8.6	8.4	8.2	8	7.8	7.6	7.4
Launch revenue	51	91	127	129	112	115	118	122	125	129	134	139	144	150
Paid readiness	0	16	19	22	25	29	33	38	43	50	57	66	76	87
Extras	1	1	1	1	2	2	2	2	2	2	3	3	3	4
Revenue	**52**	**109**	**147**	**152**	**138**	**145**	**153**	**161**	**171**	**182**	**194**	**208**	**223**	**241**
Gross profit	-17	13	37	43	35	39	44	50	56	63	71	80	91	103
Fixed cost and site fees	-60	-65	-74	-78	-78	-78	-78	-78	-78	-78	-78	-78	-78	-78
Operating profit (EBIT proxy)	**-77**	**-52**	**-37**	**-35**	**-44**	**-39**	**-34**	**-29**	**-22**	**-15**	**-7**	**2**	**13**	**24**
Cumulative	-77	-129	-165	-201	-244	-284	-318	-347	-369	-385	-392	-390	-377	-353

Red flights: more flights from Andøya than its proposed capacity (pad A plus half of pad 2, RQ3). Launch cost = Rocket Lab's cost of revenue per launch (incl. allocated production overhead) × size factor, rising by 1 ÷ 70% for every halving of cadence below 21, flat above. Fixed cost = R&D and overhead plus site fees (Nova Scotia from May 2029, pro rata). Depreciation of plant and pads is not modelled. RQ3's site load converts all field revenue into launches; here readiness is mostly a fee and failed launches are re-flown, so the launch counts differ. SQL: v_scenarios.

The three scenarios

Bear ● Medium confidence

Only today's flown demand (RQ1/RQ2) without the shift away

ASCENT (base) ● Low confidence

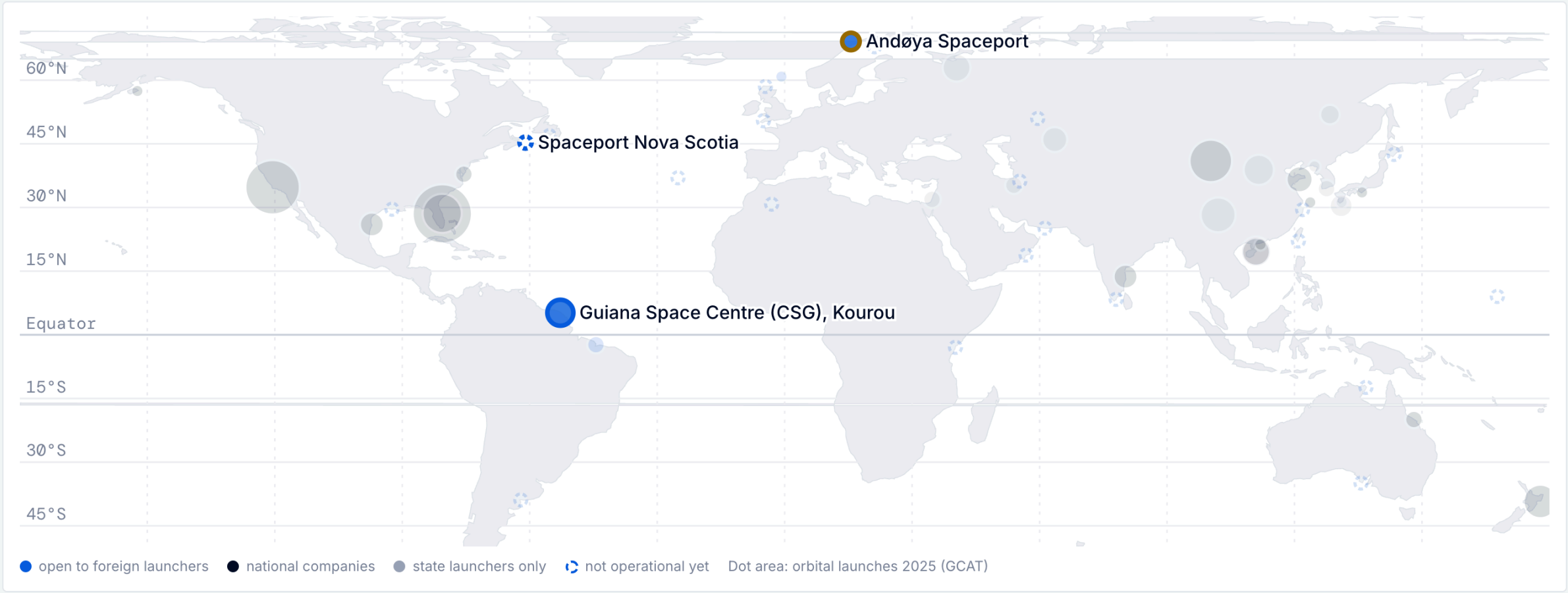
Visible demand plus two frame-contract launches a year from

Bull ● Low confidence

European space-security budgets turn into contracts:

HIGHLIGHT ON THE MAP

Nothing | **Same rockets as Andøya Spaceport** | Same rocket class (small)



Spectrum also flies, or is planned, from 2 other spaceports: Guiana Space Centre (CSG), Kourou (planned); Spaceport Nova Scotia (contracted)

NORWAY · ANDØYA SPACE

operational | Open to foreign launchers | ● High confidence

Andøya Spaceport

NORWAY · ANDØYA SPACE

operational | Open to foreign launchers | ● High confidence

Andøya Spaceport

Latitude · reach	Orbital launches 2025	Payload to orbit 2025	Launch cap per year	Heaviest vehicle, LEO
69.1° · 90–110.6°	1	–	30	1 t
Polar and SSO directly	2024: 0 · 2026 so far: 1	no satellites in 2025	CAA Norway permit for the whole spaceport (full build-out), max 4 launches at night (23–07) · 3% used in 2025	SSO 700 kg · GTO –

Vehicle	Provider	Status here	Orbits	LEO	SSO	GTO	Since	Source
Spectrum small	Isar Aerospace (DE)	active	SSO, polar	1 t	700 kg	–	2025	● Wikipedia

Year (GCAT)	Launches	Failures	Payload	to LEO	to SSO	to GTO/GEO	other	Launched by
2025	1	1	–	0	1	0	0	Isar Aerospace
2026 (to Oct)	1	0	–	0	1	0	0	Isar Aerospace

Expansion Second orbital pad in Norway's 2027 budget proposal (~NOK 650m, state-owned, leased to Andøya Space; MoD and Trade co-fund); tenant not named. Astrobotic Xodiac (reusable test vehicle, not orbital) term sheet Aug 2025, campaigns from 2026.

Isar Aerospace Exclusive pad (Orbital Launch Pad A) for up to 20 years since April 2021; Spectrum F1 failed 30 Mar 2025, F2 reached orbit 5 Sep 2026

Note Licence for the whole spaceport: 30 launches a year, at most 4 at night; headings 280–360°. The only orbital pad is Isar-exclusive, so future tenants share the 30/yr licence. Team information: further operators want to launch from 2027/28; public sources (Oct 2026) name none signed (RFA's 2020 pad deal went dormant when it moved to SaxaVord; Astrobotic Xodiac is not orbital). Owned 90% by the Norwegian state, 10% Kongsberg. Ships in the hazard zone caused the 25 Mar 2026 abort.

Sources ● CAA Norway ● Andøya Space ● Wikipedia ● Isar · pad lease ● Space Launch Schedule ● Andøya Space ● GCAT

EXPERT BASE · ROCKETS

Who launches, and with what

330 orbital launch attempts in 2025 by 12 countries and 25 launch providers, counted launch by launch from GCAT. The catalogue below holds 107 rockets with payload, status and the spaceports they fly from.

LAUNCHES BY COUNTRY AND COMPANY

2024 | 2025 | 2026 to Oct

By country of the launch provider

United States	198	98.5% ok	2,780 t
China	93	96.8% ok	328 t
Russia	17	100% ok	63.8 t
India	5	80% ok	14.8 t
France	4	100% ok	11.4 t
Japan	4	75% ok	23.7 t
Italy	3	100% ok	4.3 t
South Korea	2	50% ok	200 kg
Australia	1	0% ok	–
Germany	1	0% ok	–
Israel	1	100% ok	300 kg
Iran	1	0% ok	–

Europe together: 8 launches (87.5% successful), 15.7 t to orbit. Bars: launches; percentage: successful;

By company or agency

SpaceX (US)	170	98.8% ok	2,690 t
CASC (CN)	73	100% ok	319 t
Rocket Lab (US)	18	100% ok	2.1 t
Roscosmos (RU)	13	100% ok	58.5 t
Galactic Energy (CN)	6	83.3% ok	1.2 t
United Launch Alliance (US)	6	100% ok	76.4 t
CAS Space (CN)	5	100% ok	4.3 t
ISRO (IN)	5	80% ok	14.8 t
ExPace (CASIC) (CN)	4	75% ok	1.2 t
Arianespace (FR)	4	100% ok	11.4 t
Khrunichev (RU)	4	100% ok	5.3 t
Mitsubishi Heavy Industries (JP)	4	75% ok	23.7 t
LandSpace (CN)	3	66.7% ok	700 kg

FACT BASE

Every number, with its source

Explained · p. 28

Snapshot of the Supabase fact warehouse from 2026-10-22 11:45 UTC: 83 facts, 33 assumptions, 242 sources and 7,361 satellites. Where sources disagree, all values are kept; ✓ marks the one the tool uses.

Search facts, assumptions, sources...

☐ Only values in use Source reliability ● high ● medium ● low

Facts (83)

	Metric	Year	About	Value	Confidence	Source	Note
✓	Agreement signed (1 yes, 0 no)	2026	Canada–US Technology Safeguards Agreement	0 flag	● High confidence	● SpaceQ TSA 2026	Negotiations concluded Aug 2024, still unsigned on 5 Oct 2026: US-built satellites cannot yet launch from Canada
✓	Order backlog	2025	Rocket Lab	$1.8 bn	● High confidence	● Rocket Lab 10-K	Of which launch 475.6 M USD
✓	Cost per launch	2025	Rocket Lab	$4.8 M	● High confidence	● Rocket Lab 10-K	Electron, 21 launches in 2025: −31 % against 2023 at 2.1× the volume
✓	Cost per launch	2024	Rocket Lab	$5.7 M	● High confidence	● Rocket Lab 10-K	Electron, 16 launches in 2024; management names efficiencies of scale
✓	Cost per launch	2023	Rocket Lab	$7 M	● High confidence	● Rocket Lab 10-K	Electron, 10 launches in 2023
✓	Value of a satellite data contract	2025	Rheinmetall ICEYE	€1.7 bn	● High confidence	● ICEYE	SPOCK 1: SAR data as a service for BAAINBw, end 2025 to end 2030 with options; satellites owned by the joint venture, so the

PART 2

The case behind the screens

Three pages on what a screenshot cannot show: the argument in one page, the economics, and how we worked.

1 Answer p. 26

2 Economics p. 27

3 How we worked p. 28

4 Team p. 29

Do not commit on today's evidence: reopen diligence at three dated gates

SITUATION

Isar flies Spectrum from Andøya, its only spaceport, and looks for more production and launch sites.

COMPLICATION

From Andøya it reaches 9.3 % of the satellites launched but only 0.66 % of their mass; demand, not capacity, sets the limit.

QUESTION

Should SSVE back Isar's expansion now, and on which strategy?

ANSWER

Not on today's evidence. The strategy is right (ASCENT), the entry point is not yet: even if it works, operating break-even comes only in 2038. Reopen diligence at three dated gates.

1 From Andøya, Isar serves a polar niche: demand, not capacity, is the limit — 422 of 4,521 satellites reachable; Isar wins about 13 (€10 M) a year — p. 7 →

2 Pursue five of 13 growth fields, chosen on EVA proxy and strategic fit: grow through contracts — €88.1 M revenue and +€28.8 M EVA proxy in 2030 from the fields we recommend, before the €60 M shared R&D and overhead; four on the no-go list — p. 9 →

3 One plant, one core spaceport, one sovereign option, and a fallback outside the Arctic — The plant stays under a third full; Nova Scotia only with an anchor, Kourou if Andøya closes — p. 11 →

4 The economics do not carry the fixed cost before the late 2030s — 2030: €43 M gross profit − €78.4 M fixed cost and site fees = −€35 M; €392 M cumulative operating loss before break-even in 2038 (€445 M and 2039 with the Kourou fee) — p. 27 →

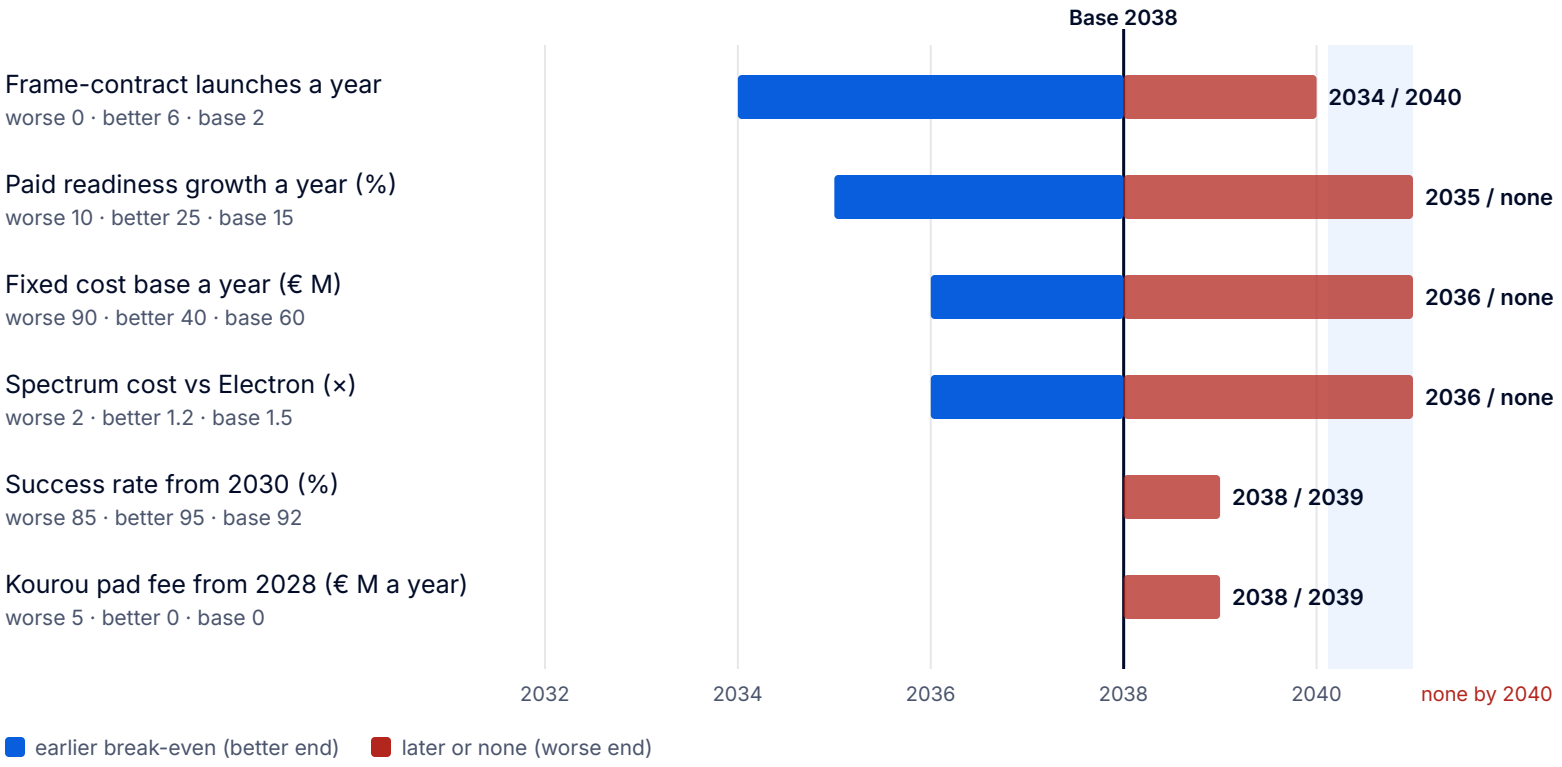
5 Three dated gates turn the no into a plan — 2027: 3 of 4 flights reach orbit · by 2029: a European state pays for launch readiness · by 2029: frame contracts for at least 2 launches a year beyond today's demand. The gates prove the base case; the contract book decides the yes (6 a year: break-even 2034) — p. 3 →

Base case ASCENT, real 2026 euros, operating result as EBIT proxy before depreciation. Source: fact warehouse (SQL views v_q1_funnel, v_q2_expansion_fields, v_scenarios), team analysis.

See it in the tool → p. 3

Even if ASCENT works, break-even comes only in 2038: signed contracts and paid readiness move it most

Exhibit 1 · Operating break-even year with one lever at each end of its range, everything else at base (labels: better end / worse end)



KEY TAKEAWAYS

- **Base case 2030:** €152 M revenue (€88.1 M recommended fields + €24 M frame contracts + €36 M Nova Scotia anchors + €4.1 M commercial), −€35 M operating result; ESA ends in 2030, −€44 M in 2031
- **€392 M** cumulative operating loss before break-even in 2038; €445 M and 2039 with the Kourou fee
- **Contracts:** 6 frame-contract launches a year instead of 2 bring break-even to 2034
- **Costs:** €90 M fixed cost, or Spectrum at 2.0× Electron's cost, mean no break-even by 2040

IMPLICATION FOR SSVE

Ask for signed frame contracts and a paid readiness deal, not for a second pad: they move the verdict most.

Operating result: EBIT proxy before depreciation; €392 M is before capex and financing, not a funding need. Source: SQL view v_scenarios; levers re-run on the tool's model, ranges from the assumption register.

See it in the tool → p. 16

Agent-first, but no AI slop: agents built and checked, the team decided, the tool runs no AI

The same inputs always give the same figures, and every figure traces back to SQL and a source.

The pipeline, and when we built each part: 48 commits in 24 hours

Sources

242 sources; GCAT: 7,361 satellites, 838 launches

`Python pipeline`

- `21 Oct 16:15` demand bottom-up from GCAT
- `18:36` expert Q&A notes
- `22 Oct 07:36` register of 56 spaceports

→

Fact warehouse

83 facts, 33 assumptions, 13 tables, 15 SQL views

`Supabase · Postgres`

- `22 Oct 06:42` schema and sourced seeds
- `08:42` Q2 strategy ASCENT
- `11:12` Q2 growth fields
- `11:40` Q3 site portfolio

→

Snapshot

CSV mirror in git, typed JSON for the tool

`uv · psycopg`

- `22 Oct 06:42` export to CSV and JSON

→

Model

pure functions for Q1–Q4 and the P&L; 31 tests

`TypeScript 5.6 · Vitest 2.1`

- `21 Oct 15:08` orbit and funnel model, first tests
- `22 Oct 13:43` P&L scenarios
- `14:04` Q4 shock

→

Tool

8 chapters and the Present page in one offline file

`React 18.3 · Vite 5.4`

- `21 Oct 15:08` scaffold
- `22 Oct 07:03` Q1 on the warehouse
- `11:41` Q2 and Q3 chapters
- `15:03` final build

Facts before features

- Every model number comes from the warehouse, with its source
- Each of the 33 assumptions has a named owner (Nils 15, Benedikt 11, Luca 7), a range and a reason
- The GCAT pipeline reproduces McDowell's published satellite totals

Business logic before UI

- Funnel, capital charge and break-even live in tested model code and SQL views, not in the screens
- The TypeScript model is checked against the SQL views
- We did not tune assumptions to force an earlier break-even

Two models, one verdict

- **claudex-loop:** Claude drafts a plan, Codex attacks it read-only, Claude revises until it holds
- 30 Codex findings on the P&L plan in three rounds, then a review of the finished code
- Storyline and verdict are the team's

Granola short summaries of our working sessions, for the project overview

Wispr Flow prompts dictated, not typed

Claude Code builder, research agents in parallel

Codex red team via claudex-loop

Source: repository (warehouse/seed, schema.sql, dashboard/package.json, git log of phase/final, progress/present-review-log.md). Logos are trademarks of their owners.

See it in the tool → p. 24

THE TEAM

Thank you for the case. We are happy to walk you through the live tool.

Our recommendation to SSVE: do not commit on today's evidence; reopen diligence at three dated gates.

Benedikt Supper

Investor view: global launch demand, competition, business case

Luca Marie Repa

Company view: Isar Aerospace's capacity, customers and plans, from public sources

Nils Wagner

Technology view: spaceflight, launch sites, orbital physics; built the tool and the fact warehouse

CODE AND DATA

github.com/wagner-nils/stern-hackathon

Public repository: the tool, the fact warehouse with its sources, the model with tests and the wiki behind the case. The README explains how to run the tool locally.

Case exercise prepared for Stern Stewart & Co. at the SSCO AI Hackathon, 21–22 Oct 2026. Figures are team estimates from public sources and the expert Q&A, not Isar Aerospace data and not investment advice. Contains participant-only material: please ask before sharing outside Stern Stewart.